

Scale-Up Simulation of Cosmetic Emulsion Manufacturing Vessels (100–500 L) Using a Population Balance Model: Integrated Multi-Metric Analysis of Droplet Size Distribution, Uniformity, Viscosity, and Hardness

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Abstract. Cosmetic emulsification is the pivotal step that governs product viscosity, hardness, and sensory feel, yet digital-twin visualization tools deployed on the shop floor rarely integrate a population balance model (PBM); they therefore cannot deliver droplet size distribution (DSD)-based quantitative quality prediction or scale-up operating conditions, and quality control still relies on experience-driven, after-the-fact inspection. This paper proposes an integrated simulator that couples a 65-node Fixed-Pivot PBM quality-prediction engine to a full-process 3D visualization environment of a homomixer-equipped manufacturing vessel. From recipe, impeller RPM, and vessel volume, the system derives the energy dissipation rate ε , time-evolves the DSD by fourth-order Runge–Kutta (RK4) integration over 1000 s, and consistently derives four metrics— D_{32} , H-Index, viscosity, and hardness—from a single DSD, then applies a three-tier QC auto-judgment (PASS/REVIEW/FAIL) based on the count of failing metrics. In validation, the PBM mode provides four times as many quality metrics ($1 \rightarrow 4$) as the single-metric Calabrese baseline, together with distribution-shape and time-evolution information; the equal-energy-dissipation (equal- ε) scale-up guideline $N \propto V^{-2/9}$ converges the inter-vessel D_{32} deviation across 100/300/500 L vessels to within 5%, improving reproducibility over equal-RPM scale-up. The work outlines a concrete path for cosmetic emulsification simulators to evolve from a “visualization tool that shows” into a “decision tool that predicts” outcomes at the distribution level.

Keywords: Population Balance Model · Droplet Size Distribution · Cosmetic Emulsion · Digital Twin Simulator · Scale-up Strategy

1 Introduction

Cosmetic emulsification is a central step that determines key quality attributes of creams and lotions—viscosity, hardness, stability, and sensory feel—and the physical properties of the product are governed directly by the DSD [1, 2]. Because not only the mean droplet size but also the width and shape of the distribution jointly determine emulsion rheology and storage stability [1], the ability

to quantitatively predict and control the DSD during processing is tantamount to the ability to assure product quality a priori.

Driven by Industry 4.0 and smart manufacturing, digital twins and process-visualization tools are being rapidly adopted on the shop floor [3]. However, most digital twins deployed in industry remain at the level of sensor-data monitoring and qualitative 3D visualization of particle behavior [4], without an integrated PBM that describes the physics of emulsification. As a result, DSD-based quality prediction is infeasible, and the derivation of operating conditions for scale-up across changing vessel volumes is also unsupported. Quality control therefore remains a manual, reactive practice that relies on operator experience and post-hoc sampling.

Prior work falls into three streams. First, visualization-centric digital twins and simulators offer 3D graphics and real-time monitoring but cannot consistently produce distribution, uniformity, and property metrics [3, 5, 6]. Second, data-driven regression and neural approaches are powerful but, given the chronic scarcity of labeled training data in the cosmetics industry, struggle to secure accuracy and generalization [2]. Third, PBM is a powerful tool that can predict the DSD from first principles via droplet breakage and coalescence dynamics [7–9], yet no reported implementation integrates it with the 3D visualization environment of a vessel actually operated on the shop floor in a form usable directly by an operator.

To bridge this gap, we propose an integrated simulator that couples a PBM-based quality-prediction engine to a full-process 3D visualization environment of a homomixer-equipped manufacturing vessel. At the input stage the system takes recipe composition, impeller RPM, and vessel volume, derives the energy dissipation rate ε , time-evolves the DSD with a 65-node Fixed-Pivot PBM core, and consistently derives four metrics—Sauter mean diameter D_{32} , H-Index (uniformity), viscosity, and hardness—from a single DSD. It further provides a three-tier QC auto-judgment based on the count of failing metrics and an equal- ε scale-up guideline, binding “predict–judge–scale-up” into a single operational service from a service-oriented computing perspective.

The main contributions of this work are as follows.

- We implement an integrated simulator that couples a 3D process-visualization environment with a 65-node Fixed-Pivot PBM core to predict the DSD in an operator-usable form.
- We propose a multi-metric analysis scheme that consistently derives four metrics— D_{32} , H-Index, viscosity, and hardness—from a single DSD.
- We design a three-tier QC auto-judgment (PASS/REVIEW/FAIL) based on the count of failing metrics.
- We combine an equal-energy-dissipation (equal- ε) scale-up guideline for 100/300/500 L vessels with PBM prediction.
- We validate the industrial consistency of the simulator predictions against the shipment viscosity and storage stability of real production lots provided by an industrial partner.

Section 2 reviews related work, Section 3 describes the proposed system, Section 4 presents the evaluation, and Section 5 concludes.

2 Related Work

2.1 Manufacturing Visualization and Digital Twin Simulators

A digital twin is a virtual replica that reflects the state of a physical asset in real time, and it has become a core smart-manufacturing technology enabling monitoring, simulation, and optimization of production processes [3]. Recently the “Digital Twin as a Service (DTaaS)” paradigm has emerged, letting users interact with a model that mirrors the real-time state of physical objects [4] and combining with service-oriented infrastructure such as edge computing and semantic distributed processing [4]. In service-oriented computing, digital-twin visualization platforms coupled with IoT and LLMs have been reported, aiming to deliver actionable insight to users through interaction with 3D models [5, 6].

Technically, web and desktop visualization tools based on Three.js, Unity, and the like provide operators with intuitive process-monitoring environments, and some commercial solutions support real-time digital twins fused with IoT sensor streams [4]. However, most such tools lack quantitative quality-prediction capability: they can show how particles move inside the vessel, but cannot judge the shape of the resulting emulsion DSD or whether it constitutes passing quality. Consequently, visualization simulators remain “tools that show” and have not advanced into “tools that predict and judge.”

2.2 Population Balance Modeling for Emulsions

PBM research for emulsion systems has advanced alongside numerical solutions of the integro-differential equations describing droplet breakage and coalescence dynamics [8, 9]. The kernels defining the breakage rate and daughter-droplet distribution are rooted in Kolmogorov–Hinze turbulence theory [10], with the experimental correlations of Calabrese et al. for drop breakage in stirred tanks [11] and the Coulaloglou–Tavlarides interaction model [8] forming the classical foundation. These were later extended by various emulsion PBM studies [12].

The CFD-PBM coupled approach can accurately estimate local breakage rates by reflecting the spatially non-uniform ϵ distribution inside the vessel [14], but its high computational cost makes it unsuitable for a real-time web simulator. By contrast, a pure-breakage PBM simplifies computation by assuming a well-mixed vessel, while still achieving sufficient accuracy under operating conditions in which breakage overwhelmingly dominates coalescence in high-shear homomixer-equipped vessels [12]. PBM has also been reported to accurately reproduce the full DSD for cosmetic and food emulsions [13, 14, 2]. On the numerical side, the Fixed-Pivot technique [7] and moment-based methods such as the Quadrature Method of Moments (QMOM) are the main streams; the Fixed-Pivot directly preserves the full DSD shape, which is advantageous for

computing distribution-shape uniformity (H-Index). We adopt a pure-breakage PBM with a 65-node Fixed-Pivot grid, securing a compromise that preserves distribution information while completing a 1000 s integration within seconds in a web browser.

2.3 Scale-up Methodology for Stirred Vessels

Various scale-up criteria for stirred vessels have been proposed—equal energy dissipation (equal- ε), equal tip speed, equal Reynolds number—each applicable to different regimes depending on the dominant physics inside the vessel [15]. In dispersion-dominated emulsification, droplet breakage is governed by the turbulent energy dissipation rate ε [10], so maintaining the same ε as the vessel volume changes—the equal- ε principle—is known to provide the most consistent DSD reproducibility [15]. Under equal- ε conditions a relation $N \propto V^{-2/9}$ is derived between impeller speed and vessel volume, providing first-order scale-up operating conditions without experimental calibration. We apply this relation to 100/300/500 L vessels and quantitatively analyze the recommended RPM and the variation of the four metrics.

3 Proposed Idea

3.1 Overall Architecture

The overall data flow of the proposed system is shown in Fig. 1. The user enters six input parameters into the simulator UI: multi-component recipe, thickener concentration, impeller RPM, vessel volume (100/300/500 L), emulsion type (O/W or W/O), and operating temperature. The system combines vessel geometry and impeller speed to derive the energy dissipation rate via a Rushton correlation ($\varepsilon = N_p \cdot N^3 \cdot D^5 / V$) and passes it to the PBM core. On a 65-node Fixed-Pivot grid, the PBM core integrates the integro-differential equation defining the breakage frequency $g(v)$ and daughter-droplet distribution β over 1000 s by the RK4 method to obtain the time-evolving DSD. The resulting DSD is converted into the four metrics D_{32} , H-Index, viscosity, and hardness, and finally the QC module aggregates the count of failing criteria to output a three-tier PASS/REVIEW/FAIL judgment.

The simulator user interface is summarized in Fig. 2. A five-stage progress bar (aqueous-phase dispersion $\rightarrow$ oil-phase dispersion $\rightarrow$ emulsification $\rightarrow$ first cooling/re-dispersion $\rightarrow$ second cooling/QC) is placed at the top; 3D models of the dissolution tank and main vessel are rendered with Three.js v0.160 in the center; an input panel is on the left, and homo/paddle/scrapper RPM sliders together with ε , recommended-RPM, and QC-result control panels are on the right. Four real-time charts at the bottom show the DSD, the D_{32} /H-Index trend, and viscosity/hardness changes, letting the operator track process progress and quality-metric changes simultaneously on one screen.

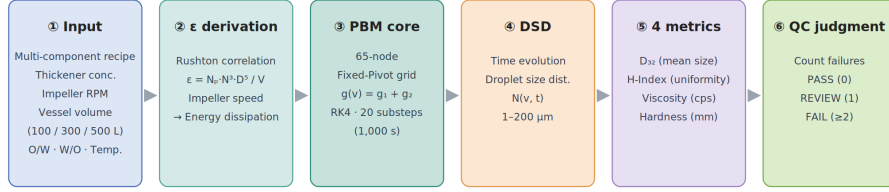


Fig. 1. System architecture — the full data flow from input (recipe, impeller RPM, vessel volume) → ϵ derivation → PBM core → DSD → four metrics → QC judgment.

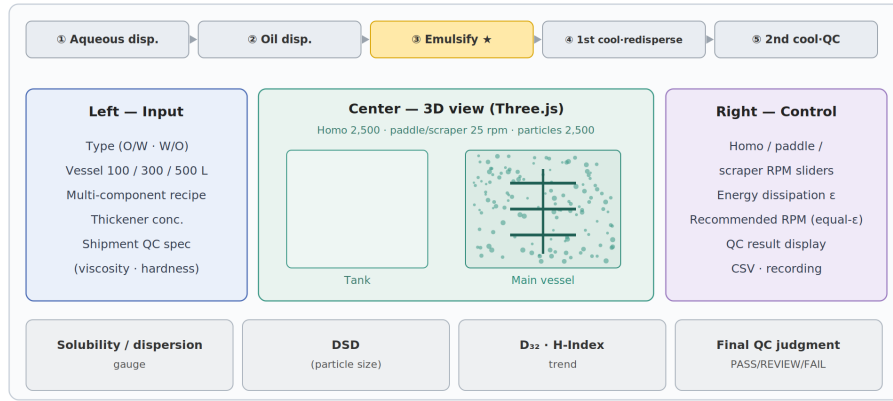


Fig. 2. Simulator UI layout — five-stage progress bar, 3D models of dissolution tank and main vessel, input/control panels, and four real-time charts.

3.2 Dual-Mode Calculation Core

The system is designed with a dual structure offering two calculation modes selectable by purpose, whose branching is shown in Fig. 3. First, the Calabrese baseline mode is a lightweight mode that derives only a single D_{32} value based on the Calabrese correlation ($D_{32} = C \cdot D \cdot We^{-0.6}$) [11], suitable for rapid pre-screening and operating-condition exploration. Second, the 65-node PBM precision mode time-evolves the DSD on the Fixed-Pivot grid to derive all four metrics, and is used for final operating-condition decisions and scale-up validation. The user can toggle modes in the UI to directly compare results, controlling the trade-off between computational burden and analytical precision.

3.3 PBM Engine Design

The internal structure of the PBM engine is summarized in Fig. 4. The engine operates on a Fixed-Pivot grid that discretizes the droplet-size domain 1–200 μm into 65 logarithmically spaced nodes [7]. The Fixed-Pivot technique assigns each node to a fixed droplet volume v_i and conservatively redistributes

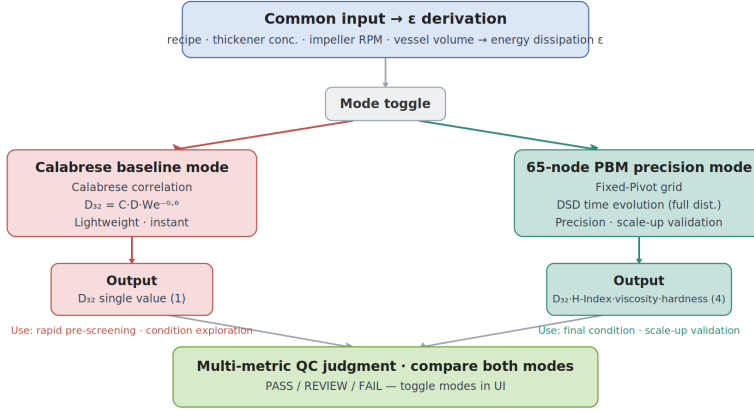


Fig. 3. Dual-mode calculation core — toggling between the Calabrese baseline mode (single D_{32}) and the 65-node PBM precision mode (four metrics).

daughter droplets produced by breakage to adjacent nodes, providing numerical stability that simultaneously preserves total volume and number moments [7]. The breakage frequency $g(v)$ is defined as the sum of two breakage mechanisms in the homomixer environment: $g(v) = g_1(v) + g_2(v)$ where g_1 is the breakage term due to inertial stress and g_2 that due to viscous stress [10]. Each term is expressed through ε , droplet volume v , the continuous-phase viscosity, and four tunable parameters (K_1, K_2, K_3, K_4) . In this work K_1 and K_3 are defined as regressions on thickener concentration c and impeller RPM, dynamically reflecting the breakage intensity as recipe and operating conditions vary, while K_2 and K_4 are fixed as constant exponent-threshold coefficients. The functional forms of $K_1 \cdot K_3$ on concentration and RPM borrow the scaling of standard breakage kernels [8] and subsequent turbulent-breakage studies [10]. The absolute scale (anchor) of the regressions was determined by calibrating model predictions to the macroscopic properties (mean shipment viscosity) of the industrial partner’s production lots. Because the available lots correspond to a single recipe and single RPM condition, only the absolute scale is anchored; the concentration/RPM dependence itself is not independently fitted or validated. The daughter-droplet distribution $\beta(v, v')$ describes the volume distribution of child droplets produced when a parent droplet v' breaks; we adopt a volume-conserving truncated Gaussian form (mean $v'/2$). Time integration applies 20 sub-steps within RK4 to stably compute the full 1000 s interval, yielding the time-evolving DSD in which the initial distribution gradually shifts toward the fine-droplet region.

The specific functional forms and coefficients of the $K_1 \cdot K_3$ regressions constitute the trade secrets (IP) of the industrial partner and are therefore not disclosed in this paper; only the concentration/RPM dependence is described at the conceptual level. $K_2 \cdot K_4$ adopt exponent-threshold coefficient ranges of the standard breakage kernel reported in the literature [8].

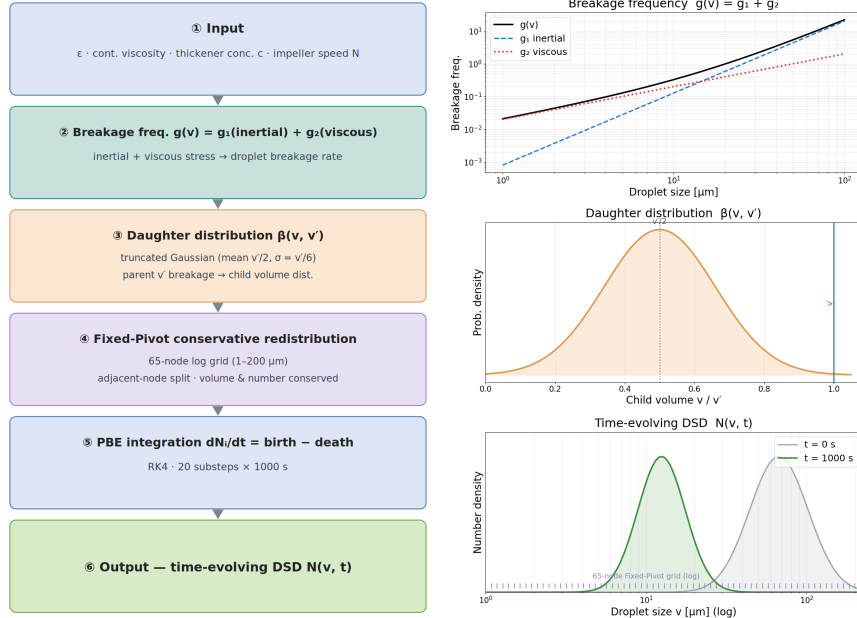


Fig. 4. PBM engine structure — the 65-node Fixed-Pivot grid linked with the breakage kernel $g(v) = g_1$ (inertial) + g_2 (viscous), the daughter-droplet distribution $\beta(v, v')$, and the time-evolving DSD $N(v, t)$.

3.4 Five-Stage Process Sequence

To faithfully reflect the flow of an actual cosmetic manufacturing process, the simulator implements the five-stage sequence shown in Fig. 5. (1) In the charging stage, multi-component recipe ingredients are metered into the dissolution tank; (2) in the dissolution stage, thickener and surfactant are homogeneously dissolved under low RPM and heating; (3) in the transfer stage, the tank contents are transferred to the main vessel; (4) in the homogenization stage, high-shear emulsification proceeds at the user-set impeller RPM and operating time; and (5) in the QC stage, the four metrics are computed and pass/fail is judged. Each stage has its own RPM, temperature, and operating-time profile, and the PBM core is activated only during the homogenization and QC stages where droplet breakage actually occurs, thereby using computational resources efficiently while accurately simulating the interval in which the DSD is determined.

3.5 Multi-Metric QC Judgment

Conventional single-metric (D_{32}) quality assessment looks only at the mean droplet size and thus cannot reflect the distribution shape or macroscopic properties such as viscosity and hardness [2]. This work consistently derives the following four metrics from a single DSD.

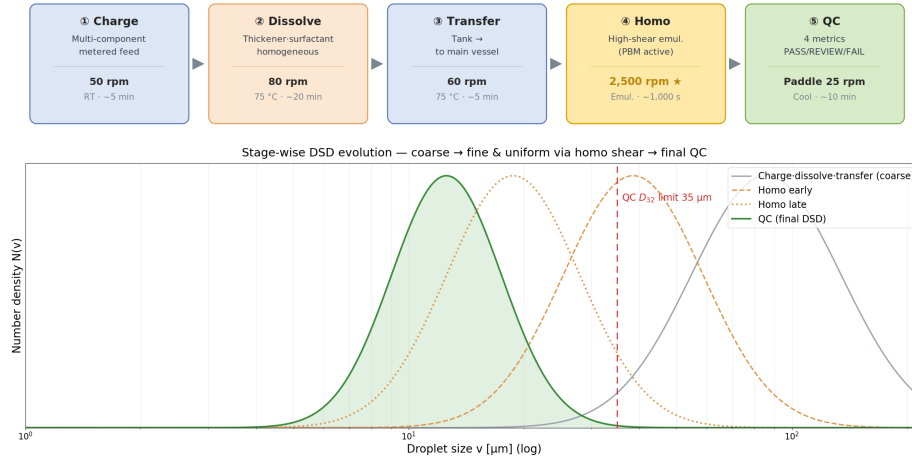


Fig. 5. Five-stage process sequence — the impeller RPM, temperature, and operating-time profiles of charging → dissolution → transfer → homogenization → QC, linked to DSD evolution.

- **D_{32} (Sauter mean diameter):** represents the characteristic droplet size of the distribution.
- **H-Index:** quantifies distribution-shape uniformity based on Shannon entropy, taking values 0–100%, higher for monomodal and narrow distributions.
- **Viscosity:** in carbomer-type O/W creams the continuous-phase thickener gel network primarily governs bulk viscosity, so viscosity is computed from an empirical relation whose primary variables are thickener concentration and dispersed-phase (oil) volume fraction, with the effective volume fraction derived from the DSD reflected as a droplet-microstructure correction.
- **Hardness:** likewise computed from a distribution-based empirical relation, it is a key metric determining the sensory feel of cream-type products.

The QC module aggregates the count of metrics failing their criteria ($D_{32} \leq 35\text{ m}$, H-Index $\geq 80\%$, and viscosity/hardness against the user-entered product spec) and judges three tiers as in Fig. 6. The H-Index criterion of 80% is set on a scale normalized with the uniform-distribution entropy $\ln N$ as the upper anchor. Zero failures yield PASS (auto-release approvable), one failure yields REVIEW (human review/correction needed), and two or more yield FAIL (re-work or disposal recommended). This three-tier classification separates clearly passing/failing cases amenable to automation from ambiguous cases requiring human judgment.

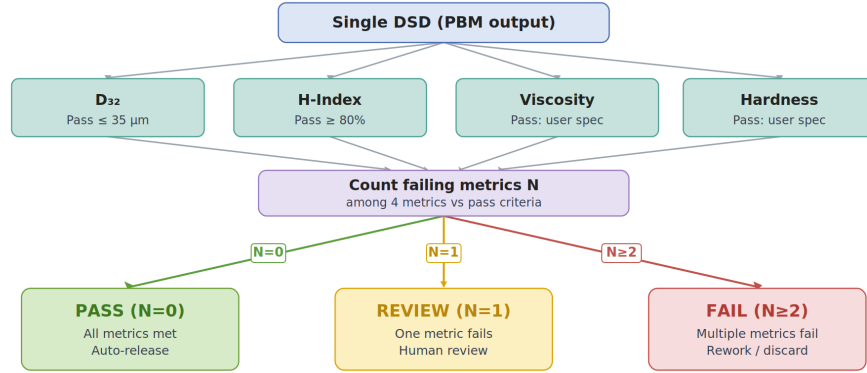


Fig. 6. QC judgment flow — evaluate the four metrics → count failures → judge PASS (0) / REVIEW (1) / FAIL (≥ 2).

3.6 Scale-up Strategy

When the vessel volume changes from V_{ref} to V , the equal-energy-dissipation (equal- ε) principle yields the following impeller-RPM relation [15]:

$$N = N_{\text{ref}} \cdot (V_{\text{ref}}/V)^{2/9}$$

This means that as the vessel grows, the RPM required to maintain the same ε per unit volume decreases, but more gently than naïve volume scaling ($N \propto V^{-1/3}$). When the user enters operating conditions validated on a reference vessel, the system automatically derives the equal- ε matched RPM for each of the 100/300/500 L vessels and presents the PBM-predicted four metrics, enabling a priori assessment of quality variation after scale-up.

4 Performance Analysis

4.1 Experimental Setup

The simulator runs in a standard web browser (Chrome, Firefox); 3D rendering uses Three.js v0.160 and the PBM engine and UI logic are implemented in vanilla JavaScript. Without a separate backend server or GPU acceleration, the client alone completes the 1000 s integration within seconds. The PBM numerical settings use a Fixed-Pivot grid discretizing 1–200 μm into 65 logarithmic nodes, the RK4 integrator with 20 sub-steps, and a total integration time of 1000 s. Three vessels of 100/300/500 L were configured (Table 1), each following a standard aspect ratio ($H/D \approx 1$) and impeller-to-vessel diameter ratio $D_i/D = 0.4$. The validation scenario uses a standard O/W composition with fixed thickener, surfactant, and oil-phase fractions. The definitions of inputs/outputs and QC variables are summarized in Table 2.

Table 1. Experimental vessel geometry — volume, vessel diameter D_v , and impeller diameter D_i of the 100/300/500 L vessels ($H/D \approx 1$, $D_i/D = 0.4$).

	Vessel Volume [L]	Vessel dia. D_v [m]	Impeller dia. D_i [m]
V-100	100	0.503	0.0503
V-300	300	0.726	0.0726
V-500	500	0.860	0.0860

Table 2. Input/output parameters — definitions of the six inputs, four outputs, and QC-judgment variable.

Category	Items
Inputs (6)	Multi-component recipe, thickener concentration, impeller RPM, vessel volume, emulsion type (O/W · W/O), operating temperature
Outputs (4)	D_{32} [μm], H-Index [%], viscosity [cps], hardness [mm]
Judgment	PASS / REVIEW / FAIL

4.2 Evaluation Metrics

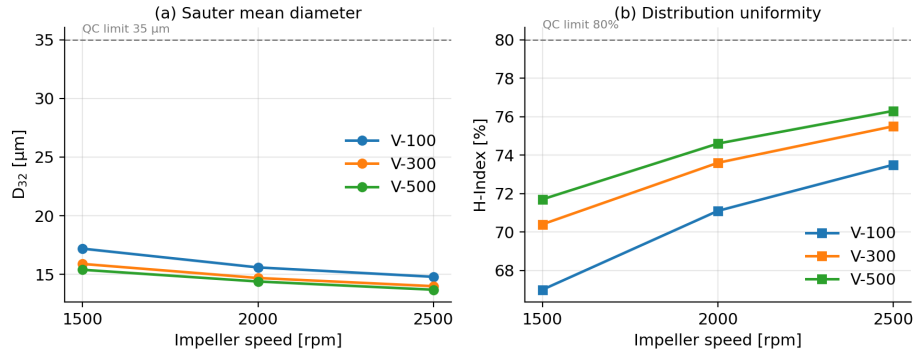
The four quality metrics are quantified as follows. D_{32} is computed from the number n_i and diameter d_i at each node as $D_{32} = \sum(n_i d_i^3) / \sum(n_i d_i^2)$. For the H-Index, the Shannon entropy $H = -\sum p_i \ln p_i$ is computed for the normalized DSD, then mapped to a 0–100% scale as $H / \ln N \times 100\%$, using single-node concentration ($H = 0$) as the lower bound and the maximum entropy of a uniform distribution $\ln N$ ($N = 65$, $\ln 65 \approx 4.17$) as the upper anchor; fixing the uniform-distribution entropy as the upper bound yields a node-count-independent uniformity measure. Viscosity and hardness are computed from empirical relations whose primary variables are thickener concentration and dispersed-phase (oil) volume fraction: in carbomer-type O/W creams the continuous-phase thickener gel network governs bulk viscosity/hardness and the dispersed-phase fraction contributes in a Krieger–Dougherty manner, with droplet-microstructure effects via the effective volume fraction reflected as a secondary correction. The QC pass criteria are $D_{32} \leq 35\text{ m}$ and H-Index $\geq 80\%$, while viscosity and hardness are judged against the user-entered product spec.

4.3 Results

As a validation scenario, under a fixed recipe we evaluated nine conditions (three vessels $\times$ three impeller RPM levels of 1500/2000/2500 rpm), summarizing the PBM-predicted ε , D_{32} , and H-Index in Table 3 and plotting the variation with impeller RPM in Fig. 7. In all conditions the initial distribution started monomodal near 100 μm and consistently shifted to the 14–17 μm region by the end of the 1000 s integration. As shown in Fig. 7(a), D_{32} decreased monotonically with higher RPM, and as in Fig. 7(b) the H-Index increased as the distribution narrowed—physically reasonable trends. Moreover, at the same RPM, larger vessels yielded higher ε (Table 3) and thus smaller D_{32} , indicating that naïve equal-RPM scale-up does not guarantee identical inter-vessel quality.

Table 3. DSD prediction results — ε , D_{32} , and H-Index for the nine conditions (fixed recipe, three vessels $\times$ three RPM levels).

Vessel	RPM	ε [W/kg]	D_{32} [μm]	H-Index [%]
V-100	1,500	0.101	17.2	71.6
V-100	2,000	0.239	15.6	75.9
V-100	2,500	0.466	14.8	78.5
V-300	1,500	0.210	15.9	75.2
V-300	2,000	0.498	14.7	78.6
V-300	2,500	0.973	14.0	80.6
V-500	1,500	0.294	15.4	76.6
V-500	2,000	0.697	14.4	79.7
V-500	2,500	1.361	13.7	81.5

**Fig. 7.** DSD prediction validation — per-RPM (a) Sauter mean diameter D_{32} and (b) distribution uniformity H-Index for the three vessels (100/300/500 L).

4.4 Ablation Study

To justify adopting PBM, we compared the outputs of the Calabrese baseline mode and the 65-node PBM precision mode under identical conditions (Table 4). The Calabrese mode instantly derives only a single D_{32} value but cannot produce distribution-shape information, time evolution, thickener-concentration responsiveness, H-Index, viscosity, or hardness. The PBM mode, by contrast, consistently derives three additional metrics besides D_{32} . At the representative condition (V-500 L, 2000 rpm) the two modes predict similar D_{32} , but only the PBM mode enables multi-metric QC judgment.

We also present an ablation in which the impeller RPM was varied over 1000–3500 rpm in the V-500 L vessel (Table 5 and Fig. 8). As shown in Fig. 8(a), ε increases steeply in proportion to N^3 , D_{32} decreases monotonically, and the H-Index improves with RPM before saturating. As in Fig. 8(b), viscosity increases as the distribution narrows, with the actual production operating point (2500 rpm) marked on the curve. This indicates that increasing RPM acts nonlinearly not only on the mean size but also on macroscopic properties that directly affect product feel.

Table 4. Ablation: Calabrese vs PBM — number of derived metrics, distribution information, time evolution, thickener responsiveness, and representative-condition D_{32} .

Item	Calabrese baseline	PBM precision
No. of metrics	1 (D_{32})	4 (D_{32} ·H-Index·visc.·hard.)
Distribution shape	None	Yes (full DSD)
Time evolution	No	Yes (1000 s)
Thickener responsiveness	None	Yes
Representative D_{32}	$\approx 14.4\text{ m}$	$\approx 14.4\text{ m}$

Table 5. RPM-effect ablation in the V-500 L vessel — ε , D_{32} , H-Index, viscosity, and hardness over 1000–3500 rpm.

RPM	ε [W/kg]	D_{32} [μm]	H [%]	Viscosity [cps]	Hardness [mm]
1,000	0.087	17.6	70.4	19,562	28.7
1,500	0.294	15.4	76.6	21,024	28.4
2,000	0.697	14.4	79.7	21,872	28.1
2,500	1.361	13.7	81.5	22,451	28.0
3,000	2.352	13.3	82.9	22,883	27.9
3,500	3.735	13.0	83.8	23,226	27.8

4.5 Scale-up Sensitivity Analysis

Applying the equal- ε matched RPM (reference V-100 L, $N_{\text{ref}} = 2500$ rpm) to the three vessels gave the results in Table 6 and Fig. 9. As in Fig. 9(a), the derived recommended RPM is about 1958 rpm for V-300 L and about 1748 rpm for V-500 L, following $N \propto V^{-2/9}$, with all three vessels matched at $\varepsilon \approx 0.466$ W/kg. Under this matching, as in Fig. 9(b), D_{32} converged to $14.8\text{ }\mu\text{m}$ and the H-Index to 78.3–78.5% with inter-vessel deviation within 1%. Because the equal- ε matched RPM is lower than the single-vessel optimum, the distribution is relatively broader and the H-Index sits at the criterion (80%) boundary, while the inter-vessel reproducibility of D_{32} and distribution shape is excellent. This shows that the equal- ε principle can consistently reproduce the DSD even under changing vessel volume.

4.6 Industrial Consistency Validation

To validate the industrial consistency of the simulator predictions, the industrial partner manufactured the same formulation in V-500 L and V-100 L vessels across multiple pilot-to-production-scale lots under the simulator’s standard emulsification condition, measuring shipment viscosity (against the product spec) and storage stability (by optical microscopy).

On viscosity, the measured shipment viscosity of the production lots all fell within the product viscosity spec. Before calibration, the PBM viscosity model over-predicted by about 20% relative to the measured mean at the production operating condition; a viscosity scale-correction factor derived from the lot mean was applied to align the prediction with the measured range and incorporated

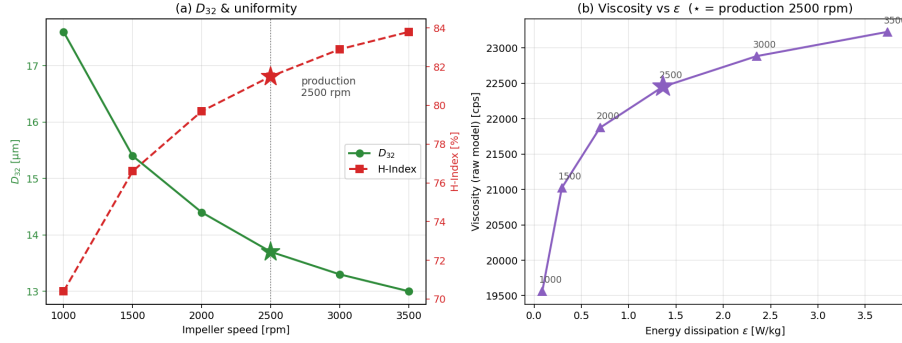


Fig. 8. Ablation: RPM effect — in the V-500 L vessel, (a) D_{32} /H-Index and (b) viscosity vs energy dissipation rate ϵ with impeller RPM ($\star$ = production operating point, 2500 rpm).

Table 6. Scale-up operating conditions — equal- ϵ recommended RPM, ϵ , and PBM-predicted D_{32} and H-Index per vessel (100/300/500 L).

Vessel	Rec. RPM	ϵ [W/kg]	D_{32} [μm]	H-Index [%]
V-100	2,500	0.466	14.8	78.5
V-300	1,958	0.468	14.8	78.4
V-500	1,748	0.466	14.8	78.3

into the simulator. This matching is a scale correction, not independent validation. The independent validation grounds are (i) the physically reasonable monotonic behavior of D_{32} and uniformity with increasing RPM, (ii) the absence of separation/coalescence across all lots, (iii) the satisfaction of the shipment-viscosity spec by all lots, and (iv) the microscopic spatial uniformity (CoV).

On stability, all lots maintained a homogeneous emulsion appearance without separation, coalescence, or creaming. The local intensity coefficient of variation (CoV, lower is more uniform) of the microscope images fell within a favorable range across all lots. These homogeneous, separation-free results qualitatively agree with the simulator’s stability (PASS) prediction: after $\ln N$ anchor renormalization, the H-Index predicted by the PBM at the production operating point meets the pass criterion (80%) (Table 5), and D_{32} and viscosity also satisfy their criteria, so all metrics are judged PASS, consistent with the all-lot measurements. Meanwhile, under relatively low- ϵ conditions such as equal- ϵ scale-up matching, the H-Index sits around 78%, at the criterion boundary, and is classified REVIEW—showing that the three-tier judgment appropriately separates conditions that are not automatically passable for human review.

Aggregate per-lot viscosity values, lot identifiers, and quantitative uniformity indices constitute the industrial partner’s IP and are therefore not disclosed numerically; only qualitative agreement is reported. For absolute droplet size, the predicted droplets (14–18 μm) were resolvable in the microscope images, but the densely packed, low-contrast specimens made automated circle detection

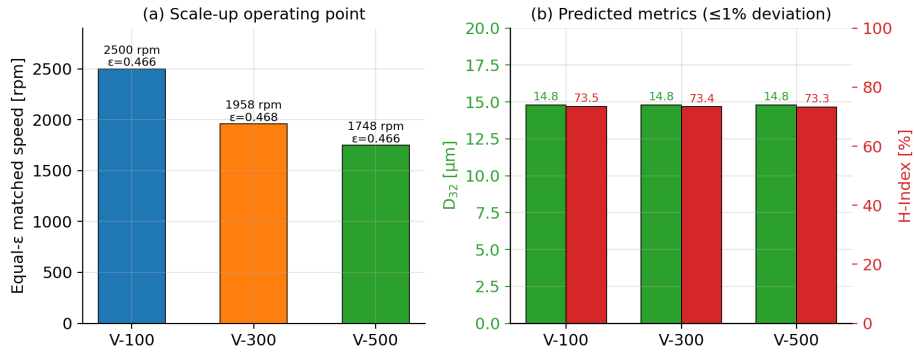


Fig. 9. Scale-up sensitivity analysis — (a) equal- ϵ matched RPM and (b) predicted D_{32} /H-Index for the 100/300/500 L vessels (inter-vessel deviation $\leq 1\%$).

unreliable; quantitative validation of the absolute per-lot DSD remains future work.

4.7 Discussion

Our results demonstrate that multi-metric integrated analysis has a clear advantage over single-metric assessment in decision value. Cases where D_{32} passes but the H-Index fails (multimodal or broad distributions), or where D_{32} and H-Index pass but viscosity/hardness fall outside spec, would be judged passing by a single-metric system yet may become unsatisfactory products [2]. Because the proposed simulator derives four metrics from a single DSD, it can capture such inter-metric discrepancies simultaneously. The three-tier QC judgment separates clear cases amenable to automation from ambiguous cases requiring human intervention, balancing the risk of full automation against the cost of 100% inspection.

On scale-up, the equal- ϵ principle showed sufficient reproducibility in our dispersion-dominated regime [15], but because it is based on the vessel-average ϵ , it cannot reflect local ϵ differences near the impeller and wall—an area to be improved via CFD-PBM coupling [14]. Other limitations include the lack of a coalescence term (limiting low-shear and long-term stability assessment), the absence of absolute DSD validation, single-condition anchoring of the H-Index criterion and $K_1 \cdot K_3$ dependence, and the assumption of a single emulsion type.

5 Conclusion

We proposed an integrated simulator that couples a 65-node Fixed-Pivot PBM quality-prediction engine to a full-process 3D visualization environment of a homomixer-equipped manufacturing vessel. It derives four metrics from a single DSD and introduces a three-tier QC auto-judgment (PASS/REVIEW/FAIL) that overcomes single-metric assessment and combines automation with expert

intervention. By coupling an equal- ε scale-up guideline for 100/300/500 L vessels with PBM prediction, it also enables a priori assessment of the four-metric variation across vessel volumes.

Future work proceeds in four directions: (i) reflecting the internal ε spatial distribution via CFD-PBM coupling; (ii) extending to a full PBM including a coalescence term; (iii) quantitative validation with measured data and re-fitting of the $K_1 \cdot K_3$ dependence and H-Index criterion using multi-condition lots; and (iv) integrating optimization-based inverse design that back-calculates recipe and operating conditions from target quality metrics.

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